INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

Data Structures and Algorithms for Problem Solving (CS1.304)

Slip Test (31 Aug 2026)

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Puzzle 1 [3 marks]

Suppose that two thieves want to rob an apartment. The first thief can carry a weight of up to **15**, and the second can carry a weight of up to **17**. The list of available items is below. How should they maximize the total value of the items they steal?

ITEM	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7
Value	24	23	18	20	13	14	11
Weight	8	7	6	5	4	4	3

Puzzle 2 [7 marks]

Devise an algorithm to solve the knapsack problem with two knapsacks of capacities $\boldsymbol{W_1}$ and $\boldsymbol{W_2}$, respectively. The number of available items is $\boldsymbol{n}$, their weights are $\boldsymbol{w_1}, \boldsymbol{w_2}, \boldsymbol{w_3}, \dots, \boldsymbol{w_n}$, and values are $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \dots, \boldsymbol{v_n}$ (all numbers are positive integers).

You shall be graded on the following parameters:

- Your code / pseudocode (possibly with comments).
- Its explanation (with a reason / logic why it works correctly).
- Its runtime analysis (the quicker, the better).

Thank You!